Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

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Real-Time Event Synchronization Engine

SDG: 9 – Industry , Innovation and Infrastructure

Presented By:

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INTRODUCTION

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The Real-Time Event Synchronization Engine is designed to provide dynamic and efficient data updates in modern web applications. Traditional systems often require manual page refresh, which leads to delays and outdated information. This project overcomes that limitation by enabling automatic updates without reloading the webpage. The system allows users to add and view events instantly, ensuring real-time synchronization. It is developed using HTML, CSS, and JavaScript for the frontend, and PHP with MySQL for backend processing. The Fetch API is used to establish asynchronous communication between client and server. This project has real-world applications in event management systems, dashboards, and notification platforms. Overall, it demonstrates how dynamic web pages can be created using database queries and real-time synchronization techniques.

PROBLEM STATEMENT

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In many web applications, event data is not updated in real time and often requires manual page refresh, leading to delays and inefficiency. Users may not see the latest updates instantly, which affects decision-making and user experience. Existing systems lack proper synchronization between the database and the user interface. This creates inconsistencies and reduces system reliability. Therefore, there is a need for a dynamic and automated system that can update event data instantly. The proposed Real-Time Event Synchronization Engine addresses this issue by enabling seamless communication between frontend and backend. It ensures that newly added events are displayed without page reload. This improves efficiency, accuracy, and overall performance of the system.

OBJECTIVE(S)

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- To design and develop a Real-Time Event Synchronization Engine using HTML, CSS, JavaScript, PHP, and MySQL for dynamic data handling.
- To implement and evaluate real-time data updates using asynchronous communication (Fetch API) to improve system efficiency and user experience.

PROPOSED SYSTEM

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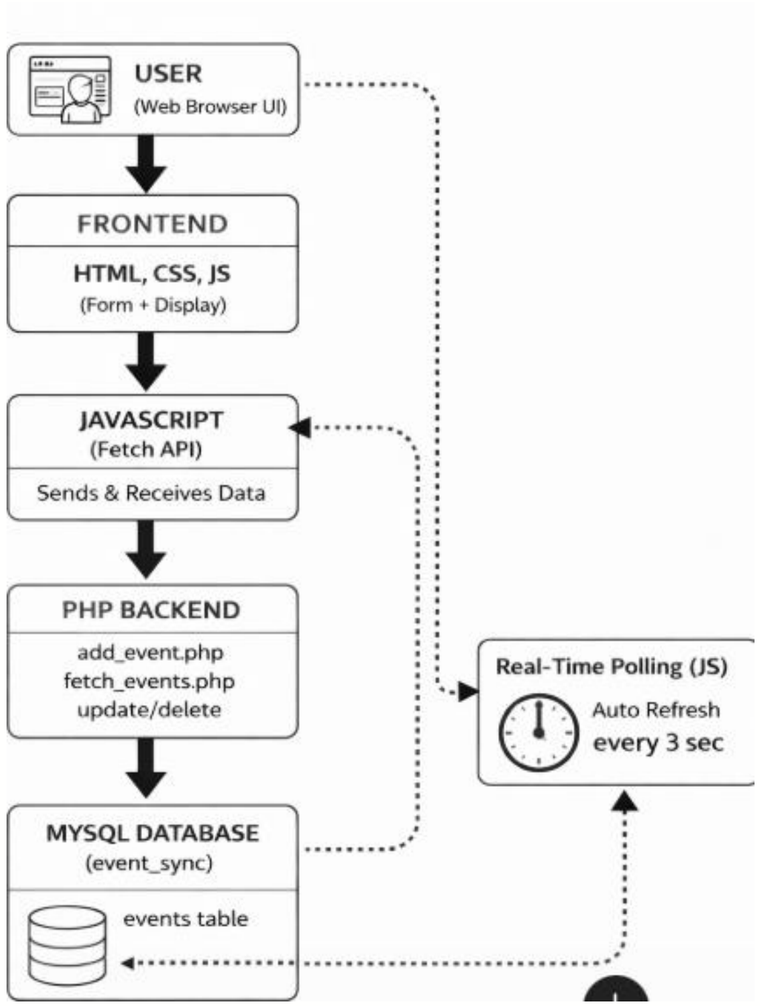
- The proposed system is a Real-Time Event Synchronization Engine that enables dynamic updating of event data
- It uses HTML, CSS, and JavaScript for the frontend and PHP with MySQL for backend data management.
- The system allows users to add, view, update, and delete events efficiently.
- It implements asynchronous communication using Fetch API to ensure real-time data synchronization.
- The system continuously fetches updated data from the database using polling techniques.
- It solves the issue of delayed updates by automatically refreshing event data on the user interface.
- Compared to traditional systems, it reduces manual effort and improves responsiveness.
- Overall, the system provides a fast, reliable, and user-friendly solution for managing real-time event data.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW

```
          [ USER ]
              |
              | Enter Event Details
              v
      ( 1.0 EVENT SYSTEM )
           /         \
          /           \
         v             v
[ Display Output ]   [ Event Data ]
                          |
                          | Store / Retrieve
                          v
                   [ MYSQL DATABASE ]
```

The user provides event details as input to the system, which processes the data.

The system stores and retrieves event data from the MySQL database.

The processed data is displayed back to the user in real time without page refresh.

TECHNOLOGIES USED

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- **Frontend:** HTML5, CSS3, JavaScript (ES6)
- **Backend:** PHP (Server-side scripting)
- **Database:** MySQL (event sync database)
- **Tools:** XAMPP (Apache Server), phpMyAdmin, Web Browser (Chrome/Edge), VS Code / Notepad++
- **Hardware/Software Requirements:**
- System with minimum 4GB RAM
- Windows OS
- XAMPP installed
- Web browser for execution

MODULE DESCRIPTION

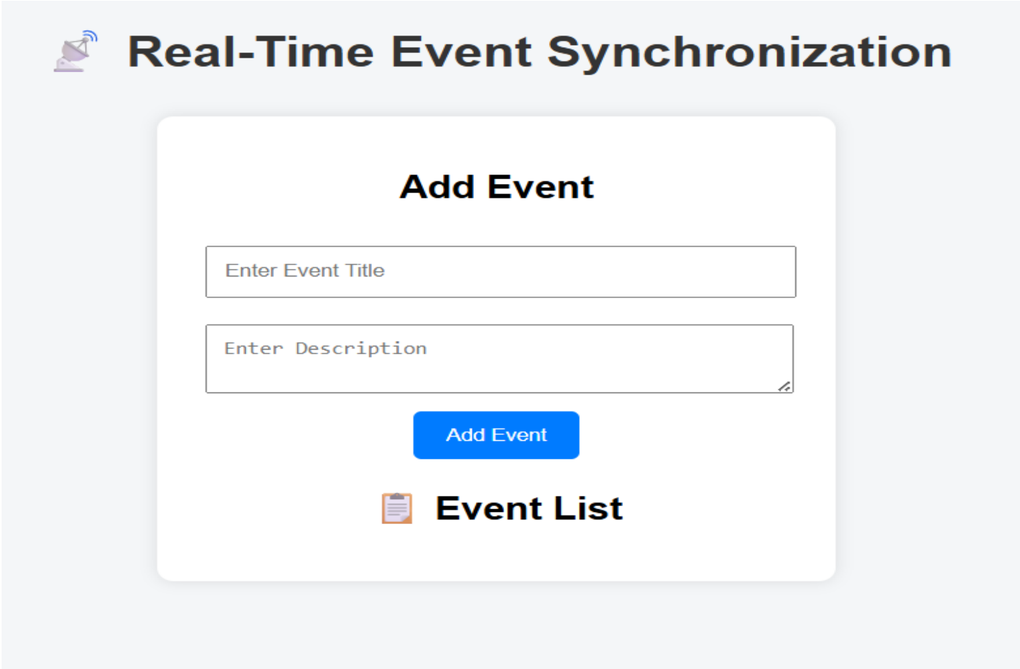
Module 1: Event Input Module
Handles user input by allowing users to enter event details such as title and description through the web interface.

Module 2: Event Processing Module
Processes user requests using JavaScript (Fetch API) and sends data to the PHP backend for further operations.

Module 3: Database Management Module
Manages storage, retrieval, update, and deletion of event data in the MySQL database ensuring data consistency.

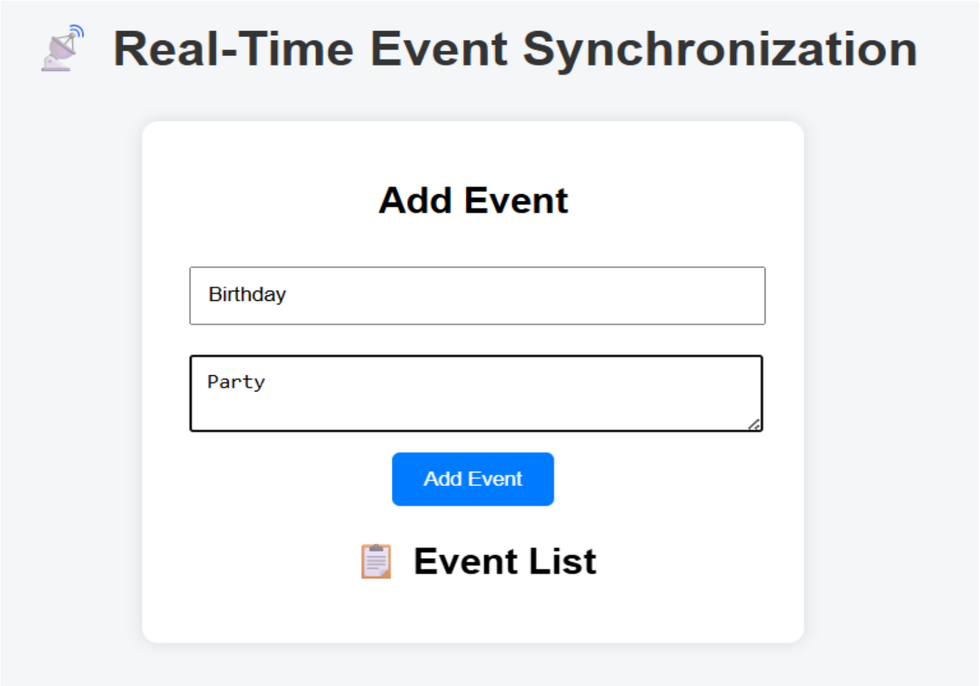
Module 4: Real-Time Synchronization Module
Fetches updated event data periodically and displays it on the webpage without reloading, ensuring real-time updates.

IMPLEMENTATION SCREENSHOTS



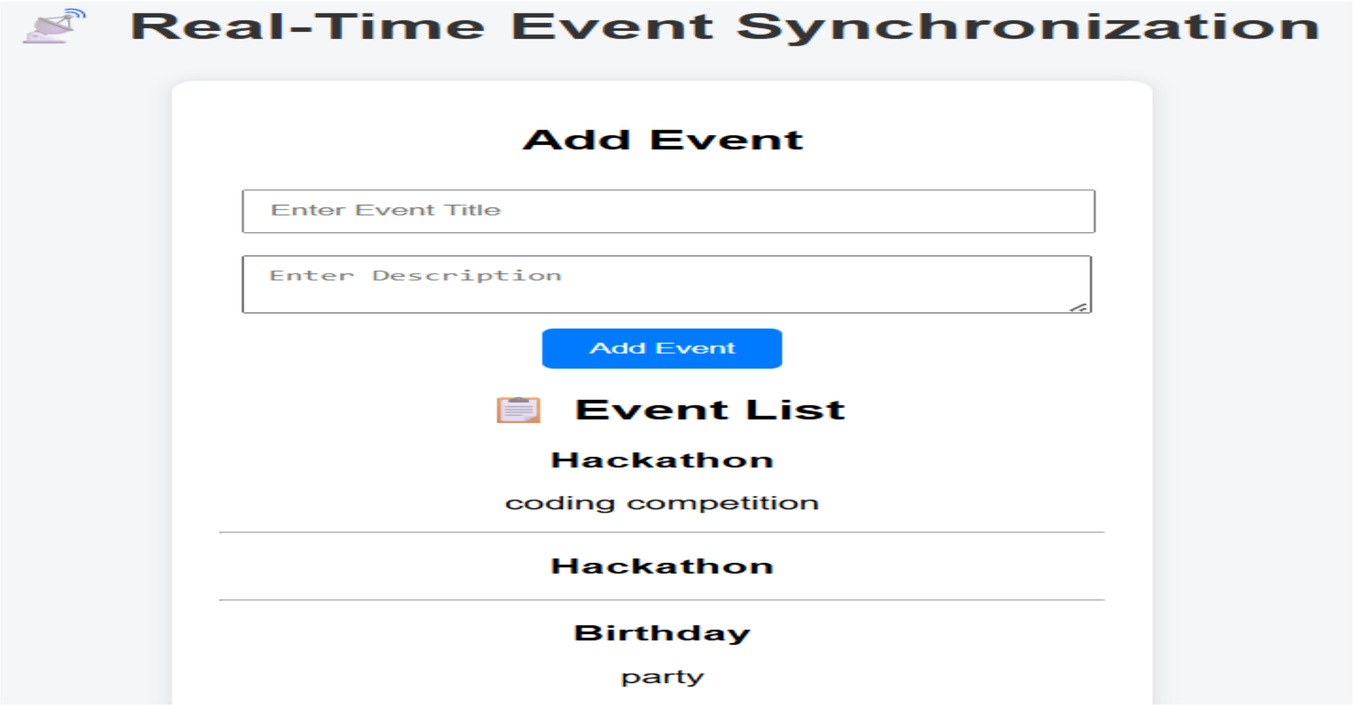
The Real-Time Event Synchronization system where users can enter event details.
It provides an interface to add events and view them dynamically in the event list section.”

IMPLEMENTATION SCREENSHOTS



This screen shows the user entering event details such as title and description in the form. After clicking the ‘Add Event’ button, the data is sent to the backend for processing and stored in the database.
The system prepares to display the updated event list dynamically.

IMPLEMENTATION SCREENSHOTS



This screen displays the list of events retrieved from the database after insertion.
The events are shown dynamically without refreshing the page, demonstrating real-time synchronization.
It confirms successful storage and retrieval of event data in the system.

IMPLEMENTATION SCREENSHOTS

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Showing rows 0 - 2 (3 total, Query took 0.0011 seconds.)

```
SELECT * FROM `events`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refres

Show all | Number of rows: All

Extra options

←T→			id	title	description	event_time
Edit	Copy	Delete	1	Birthday	party	2026-03-23 23:50:25
Edit	Copy	Delete	2	Hackathon		2026-03-24 00:00:22
Edit	Copy	Delete	3	Hackathon	coding competition	2026-03-24 00:00:36

This screen shows the event data stored in the MySQL database after insertion

ADVANTAGES & LIMITATIONS

ADVANTAGES

•Provides real-time updates without reloading the webpage.

•Reduces manual effort and improves user experience.

•Simple and easy to use interface for adding and viewing events.

•Efficient data storage and retrieval using MySQL.

•Uses asynchronous communication, making the system fast and responsive.

LIMITATIONS

- Uses polling (every few seconds), which may increase server load.
- No user authentication (anyone can add/delete events).
- Limited scalability for large number of users.
- Basic UI without advanced features or security.

Thank you